Interpreting Correlation and Least-Squares Regression

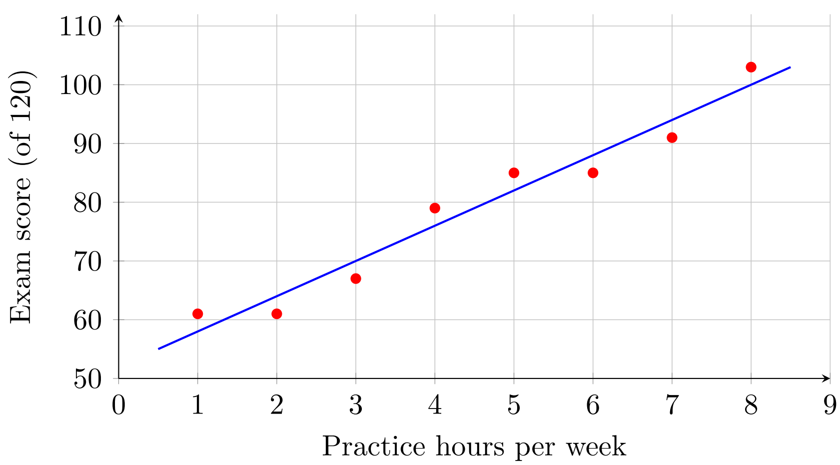
Predicting from the least-squares line, residuals, reading r and r-squared, and knowing when not to trust the model.

A study skills coordinator recorded, for eight students, the hours of practice problems done per week (x, in hours) and the score on a 120-point end-of-unit exam (y, in points). Statistical software returned the least-squares regression line

$$\hat{y} = 52 + 6x \qquad \text{with correlation } r = 0.977.$$

Every problem below uses this data set and this line. Round any percent to one decimal place.

Practice hours x	1	2	3	4	5	6	7	8
Exam score y	61	61	67	79	85	85	91	103



1. Use the least-squares line $\hat{y} = 52 + 6x$ to predict the exam score, in points, of a student who does 5 hours of practice problems per week.

Answer: ____________________

2. Read the table. What exam score did the student who practiced 5 hours per week actually earn, in points?

Answer: ________________

3. Find the residual for the 5-hour student: the observed score of 85 points minus the predicted score of 82 points. Then say in one sentence whether the line under-predicted or over-predicted that student.

Answer: ________________

4. The software reported the correlation $r = 0.977$. Compute the coefficient of determination r^2, rounded to three decimal places, and state what percent of the variation in exam scores the model accounts for.

Answer: ________________

5. Use the line to predict the exam score, in points, of a student who does 8 hours of practice problems per week. Compare your prediction with the observed score of 103 points in the table.

Answer: ________________

6. Look at the scatter plot. Count how many of the eight data points lie *above* the regression line, then write that count as a fraction of all 8 points, in lowest terms. Explain in one sentence why a least-squares line is expected to split the points roughly this way.

Answer: ________________

7. Find the mean exam score, in points, of the eight students. Then substitute the mean practice time $x = 4.5$ hours into $\hat{y} = 52+6x$ and compare. What property of the least-squares line does this confirm?

Answer: ________________

8. The coefficient of determination for this model is $r^2 = 0.955$. What percent of the variation in exam scores does the model leave *unexplained*? Give your answer to one decimal place.

Answer: ________________

9. A student proposes practicing 20 hours per week and uses the line to predict the resulting score. Compute $\hat{y}$ at $x = 20$ hours, then state what is wrong with the number you get, given that the exam is out of 120 points.

Answer: ________________

10. Write a short paragraph, in complete sentences, answering both parts.

(a) Using your answer to problem 9, explain why the model must not be used outside the range of practice hours actually observed, from 1 to 8 hours.

(b) The correlation $r = 0.977$ is very strong. Explain why that still does not show that practice *causes* higher scores. Name one confounding variable that could produce this pattern on its own, and describe one study design that could settle the question.